Scalability & Future Plan

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Team ID	Rahul-DS633
Project Name	CrimeVision AI - Intelligent Multimodal Forensic Analysis Platform
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-camera monocular 3D depth projection lacks stereoscopic multi-angle parallax calibration.	Limits sub-millimeter metric distance accuracy in highly cluttered 3D crime scenes.	**Medium**
2	Cloud free-tier hosting (512MB RAM) requires CPU-quantized models and limits concurrent high-bitrate video streams.	Limits simultaneous real-time video processing to 1-2 concurrent 1080p surveillance streams.	**High**
3	Extreme acoustic background noise (e.g. blaring sirens, loud engines) slightly elevates Whisper speech word error rate.	May necessitate secondary manual confirmation of low-confidence witness audio segments.	**Low**

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	**User Load**	Single-tenant web dashboard serving up to 10 concurrent forensic investigators.	Containerized Kubernetes cluster deployment with Horizontal Pod Autoscalers (HPA) and Redis session caching.
2	**Data Storage**	Local ephemeral file storage (data/uploads) on application instance.	Migrate to AWS S3 / Cloudflare R2 object storage with AES-256 encryption and PostgreSQL metadata indexing.
3	**Performance**	CPU inference with 1.0 Hz video frame sampling (~1.2x video duration).	Deploy NVIDIA TensorRT GPU acceleration (T4/A10G) enabling real-time 30 FPS multi-camera inference at <100ms latency.
4	**Security**	HTTPS web endpoint with CORS restriction and session isolation.	Implement Role-Based Access Control (RBAC), CJIS-compliant audit logs, end-to-end encryption, and SHA-256 evidence hashing.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Real-time RTSP/ONVIF Live CCTV Stream Ingestion	Q3 2026	Enables continuous 24/7 automated monitoring of municipal traffic and commercial facility feeds with instant threat alerting.
Phase 2	Multi-Camera Cross-View Re-Identification (Re-ID)	Q4 2026	Seamlessly tracks suspects across multiple disjoint camera perimeters using deep visual appearance embedding vectors.
Phase 3	Edge Computing Deployment (NVIDIA Jetson AGX Orin)	Q1 2027	Enables on-premise forensic edge appliances for police cruisers and local command centers without cloud internet dependency.
Phase 4	Biometric Facial Recognition & National Database Integration	Q2 2027	Cross-references flagged suspect faces against lawful authorized criminal warrants databases with strict privacy safeguards.